# ABHISHEK JANAKKUMAR JANI

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## OBJECTIVE

Software engineer with an M.S. in Computer Science, focused on backend and distributed systems. I build services where correctness under failure matters — event-sourced ledgers, LLM proxies with measured cost ledgers, and cloud-native data pipelines — and the interfaces that make them usable. Two software engineering internships, plus a year and a half teaching programming and AI at the graduate-assistant level.

## EDUCATION

**California State University, Fullerton** — *Aug 2024 – May 2026*
*Master of Science, Computer Science | Fullerton, CA*

**Atharva College of Engineering, University of Mumbai** — *Aug 2020 – May 2024*
*Bachelor of Engineering, Information Technology | Mumbai, India*

## TECHNICAL SKILLS

**Languages:** Python, Java 21, TypeScript, JavaScript (ES6+), SQL, C, C++, Bash

**Backend:** Spring Boot 3, FastAPI, Node.js, Express.js, Django, Flask — REST API design, microservices, event-driven architecture, CQRS, transactional outbox, saga orchestration, streaming proxies

**Data & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, Snowflake, SQLite — schema design, migrations, indexing, query optimization; Apache Kafka (Redpanda) event streaming, AWS SQS, append-only event stores

**Frontend:** React, Next.js 15, Redux, Tailwind CSS, HTML5/CSS3, Streamlit — component architecture, state management, real-time data rendering

**Cloud & DevOps:** AWS (Lambda, S3, SQS, RDS / Aurora, CloudWatch, Cognito), Terraform (IaC), Docker, Kubernetes, GitHub Actions CI/CD, Jenkins, Linux

**AI Engineering:** LLM API integration and gateway proxying, Model Context Protocol (MCP) tool interfaces, agent orchestration and multi-agent workflows, context optimization, skill compilation and replay, output parity grading; Claude Code, Claude API, Cursor, GitHub Copilot

**Platforms & Integrations:** InsForge (agentic dev tooling), EverOS (agent memory — cases & skills), VoiceOS (voice-native agent control, MCP registration), a1mobile (programmatic SMS, outbound calling, OTP verification), Snowflake (cost ledger + SAVINGS views), Novita, Razorpay, AWS Textract, Streamlit, Postman

**Practices:** Unit / integration / regression testing (pytest, Jest, JUnit, Playwright), peer code review, structured logging and distributed tracing, Git, Agile/Scrum, technical documentation

## PROFESSIONAL EXPERIENCE

**Teaching Associate — Computer Science** — *Jan 2025 – May 2026*
*California State University, Fullerton | Fullerton, CA*

- Instructed **Python Programming (CPSC 223P)** and **Artificial Intelligence (CPSC 481)** for 35–40+ students per section, covering data structures, algorithms, object-oriented design, neural networks, and debugging.
- Built **automated test suites** evaluating 40+ code submissions per assignment against defined correctness criteria, and gave written review feedback on every submission.
- Authored lecture notes, lab specifications, and assignment documentation, and mentored students one-on-one through design and correctness problems.

**Software Developer Intern** — *Nov 2023 – Apr 2024*
*Branding Catalyst Pvt Ltd | Mumbai, India*

- Developed and optimized **backend services and REST APIs** with Node.js, Express, and MongoDB, improving overall system **performance by 20%**.
- Designed the core **database schema** and built aggregation pipelines with targeted indexes, cutting **data retrieval time by 25%**.
- Implemented secure **authentication** with JWT and bcrypt, reducing reported vulnerabilities by **50%**; built asynchronous CSV ingestion, AWS S3 uploads, and email/SMS delivery, and integrated the Razorpay payment gateway.
- Reduced production bugs by **30%** through Postman API testing and log-based debugging; contributed in **Agile sprints, code reviews, and sprint planning**.

**Full Stack Developer Intern** — *Mar 2023 – Apr 2023*
*Peaceinfotech Services Pvt Ltd | Mumbai, India*

- Built a full-stack **Bus Booking System** (React.js, Node.js, Express.js) exposing **REST APIs** for real-time seat availability, booking, and scheduling.
- Implemented **Redux** state management and a responsive UI, improving data-fetching efficiency by **20%** and reducing post-deployment issues by **15%**.

## PROJECTS

**Amortize — Cost-Reducing LLM Gateway Proxy** — *2026*
*Python, FastAPI, Snowflake, SQLite, EverOS, Streamlit | github.com/Abhishekjani2509/Amort*

- Built a **local proxy** that any LLM client points at with a single environment variable — no SDK swap or code change — that reduces token spend while returning field-identical output.
- Implemented context **lightening** (tool schemas compacted to stubs, oversized results spilled to disk behind handles) for a measured **28% token reduction** on a cold run at parity.
- Built a skill-compilation layer that records each run as a case, distils repeats into a replayable procedure, and re-executes it as code — **97% token reduction** on repeat runs, with a **parity grader** proving output equivalence.

- Wrote a **per-step cost ledger** to Snowflake with automatic SQLite fallback, batched writes, retry, and mid-run degradation that replays the in-flight buffer — every claimed number traceable to a ledger row.

**Reckon — Event-Sourced Distributed Ledger** *2026*

*Java 21, Spring Boot 3, PostgreSQL, Apache Kafka (Redpanda), Redis, Next.js 15 | github.com/Abhishekjani2509/reckon*

- Built a double-entry wallet ledger on an append-only **event store** where every balance is derived by replaying an immutable event log — the audit trail and the data are the same artifact.
- Implemented **CQRS**, the **transactional outbox** pattern for exactly-once event publication, and **saga orchestration** with compensating actions — all from scratch rather than via a framework.
- Streamed domain events through Kafka into Redis-backed read models serving a real-time Next.js dashboard.

**Crew — Voice-Driven Multi-Agent Orchestration** *2026*

*Node.js, Swift / AppKit, MCP | 5-person team | github.com/Abhishekjani2509/crew*

- Built the **desktop dock UI** — characters, speech bubbles, and narration pacing — rendering five parallel agents as something you watch rather than a transcript you read afterward.
- Solved **event-ordering and pacing** problems in the narration pipeline: fire-and-forget POSTs raced and delivered lines out of sequence, and pacing was set from measured speech duration rather than guessed timeouts.
- Contributed to a three-process architecture with **frozen HTTP contracts** defined on day one so five people on five machines could build in parallel without blocking each other.

**Computah — Platform-Agnostic Coding Agents** *2026*

*TypeScript, Next.js, Node.js, Playwright, MCP | 4-person team | github.com/Abhishekjani2509/Computah*

- Built an **MCP tool interface** interoperable with Claude Code and Cursor, plus a Playwright **self-verification loop** that executes generated code, captures failures, and re-attempts until correctness criteria pass.
- Implemented **real-time cross-platform session sync** across browser, Discord, and voice clients, where event ordering and state convergence between transports was the core problem.

**DocuPop — Serverless Document Processing Platform** *2026*

*TypeScript, Next.js 15, Python, AWS Lambda, S3, SQS, Aurora PostgreSQL, Terraform | Graduate capstone*

- Architected a **serverless pipeline** ingesting documents at scale and extracting structured, queryable data, decoupling ingestion from compute with SQS, retries, and dead-letter handling.
- Provisioned all infrastructure as code with **Terraform** and automated build, test, and deployment through **GitHub Actions CI/CD**; instrumented CloudWatch logs, metrics, and traces for failure tracing.